INVARIANT AFFINE OPEN CONTAINING A POINT WHOSE ORBIT HAS ONE-DIMENSIONAL LOCAL RINGS IN A FINITE GROUPOID

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- A finite set of points of codimension at most one in a separated locally Noetherian scheme is contained in an affine open (varieties chapter, Proposition varieties-proposition-finite-set-of-points-of-codim-1-in-affine, canonical tag 09NN). cited at proof lines 3035-3038. Puts the finite R-orbit of u into an affine open U', starting the reduction chain.
- Descent of invariant affine opens along integral morphisms met by every orbit (more-groupoids chapter, Lemma lemma-find-affine-integral, no canonical tag in the snapshot; no-tag pointer to the source label line). cited at proof lines 3046-3047 and 3052-3053. Justifies replacing U by its reduction and later by the closure of the invariant dense open, moving u to an orbit point.
- Existence of an invariant open W and an invariant closed W' in W with W a thickening of W', the orbit image condition on the closure of W, and finite locally free restrictions on W' (more-groupoids chapter, Lemma lemma-finite-flat-over-almost-dense-subscheme, no canonical tag in the snapshot; no-tag pointer to the source label line). cited at proof lines 3047-3049. Supplies the dense invariant open W with finite locally free structure maps used in the main construction.
- Invariant affine open containing a finite set of points for a groupoid with finite locally free structure maps (more-groupoids chapter, Lemma lemma-find-invariant-affine, canonical tag 03JE). cited at proof lines 3060-3061. Settles the case u in W directly.
- Any finite set of points of a quasi-affine scheme is contained in an affine open (properties chapter, Lemma properties-lemma-ample-finite-set-in-affine, canonical tag 01ZY). cited at proof lines 3062-3064. Input used to apply lemma-find-invariant-affine to the quasi-affine W.
- Distinct points in the same orbit of a groupoid with finite s, t are never specializations of one another (more-groupoids chapter, Lemma lemma-no-specializations-map-to-same-point, no canonical tag in the snapshot; no-tag

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pointer to the source label line). cited at proof lines 3068-3070. Ensures a point specializing to an orbit point lies outside the orbit, so assumption (3) makes it a generic point of U.

- Almost invariant functions: for a groupoid with s, t finite of finite presentation, points with generic orbits, an immersion into an affine scheme and an ideal I disjoint from the image with closed union, some element of I cuts out an invariant affine open containing the points (more-groupoids chapter, Lemma lemma-find-almost-invariant-function, no canonical tag in the snapshot; no-tag pointer to the source label line). cited at proof lines 3082-3086 and 3110-3113. Produces f, giving an invariant affine open inside W containing all xi_j, and h, giving an invariant affine open inside Z containing u; these are the two pieces glued into V.
- The norm of the pullback of a function along the source map of a groupoid is invariant (more-groupoids chapter, Lemma lemma-determinant-trick, canonical tag 03BH). cited at proof lines 3116-3117. Makes g, the norm of the pullback of h along the structure maps, an R_W-invariant function.
- In Noetherian topological spaces the closed subsets are exactly the constructible specialization closed subsets (topology chapter, Lemma topology-lemma-characterize-closed-Noetherian, canonical tag 0542). cited at proof lines 3130-3133. Reduces openness of the constructible set V to a generalization closedness check, carried out by the explicit chase.
- Constructible subsets closed under specialization characterize closed subsets, general version (topology chapter, Lemma topology-lemma-constructible-stable-specialization-closed, canonical tag 0903). cited at proof lines 3133-3135. Cited alongside 0542 as the more general form of the same openness criterion.
- Affineness criterion: a quasi-compact immersion into an affine scheme whose preimage of a principal open is affine and whose intersection with the principal closed part is closed has affine source (more-groupoids chapter, Lemma lemma-get-affine, no canonical tag in the snapshot; no-tag pointer to the source label line). cited at proof lines 3164-3166. Concludes that V is affine from the immersion into the affine scheme A_h localized.

Source reference. Original proposition 09NN: sources/stacks-chow/tags.txt, source lines 12221–12221. Cited at proof lines 3035-3038: puts the finite orbit of u into an affine open.

Source reference. Original lemma lemma-find-affine-integral: sources/stacks-more-groupoids/pinned-original-source.tex, source lines 2887–2887. Cited at proof lines 3046 and 3052-3053: reduction to reduced U and replacement by the closure of W.

Source reference. Original lemma lemma-finite-flat-over-almost-dense-subscheme: sources/stacks-more-groupoids/pinned-original-source.tex, source lines 2672–2672. Cited at proof lines 3047-3049: supplies the invariant dense open W with finite locally free restrictions.

Source reference. Original lemma 03JE: sources/stacks-chow/tags.txt, source lines 4636–4636. Cited at proof lines 3060-3061: settles the case u in W.

Source reference. Original lemma 01ZY: sources/stacks-chow/tags.txt, source lines 2613–2613. Cited at proof lines 3062-3064: input for applying 03JE to the quasi-affine W.

Source reference. Original lemma lemma-no-specializations-map-to-same-point: sources/stacks-more-groupoids/pinned-original-source.tex, source lines 2965–2965. Cited at proof lines 3068-3070: points specializing into the orbit lie outside the orbit.

Source reference. Original lemma lemma-find-almost-invariant-function: sources/stacks-more-groupoids/pinned-original-source.tex, source lines 2923–2923. Cited at proof lines 3082-3086 and 3110-3113: produces f and h, the two affine pieces of V.

Source reference. Original lemma 03BH: sources/stacks-chow/tags.txt, source lines 4348–4348. Cited at proof lines 3116-3117: makes the norm function g invariant.

Source reference. Original lemma 0542: sources/stacks-chow/tags.txt, source lines 6625–6625. Cited at proof lines 3130-3133: reduces openness of V to the generalization check.

Source reference. Original lemma 0903: sources/stacks-chow/tags.txt, source lines 11396–11396. Cited at proof lines 3133-3135: the more general form of the same criterion.

Source reference. Original lemma lemma-get-affine: sources/stacks-more-groupoids/pinned-original-source.tex, source lines 2987–2987. Cited at proof lines 3164-3166: concludes V is affine.

Lemma 0.2. *Let (U, R, s, t, c) be a groupoid scheme. Let $u \in U$. Assume*

(1) *s, t are finite morphisms,*
(2) *U is separated and locally Noetherian,*
(3) *$\dim(\mathcal{O}_{U,u'}) \leq 1$ for every point u' in the orbit of u.*

Then u is contained in an R-invariant affine open of U.

Proof. The R-orbit of u is finite. By conditions (2) and (3) it is contained in an affine open U' of U, see Varieties, Proposition 09NN. Then $t(s^{-1}(U \setminus U'))$ is an R-invariant closed subset of U which does not contain u. Thus $U \setminus t(s^{-1}(U \setminus U'))$ is an R-invariant open of U' containing u. Replacing U by this open we may assume U is quasi-affine.

By Lemma lemma-find-affine-integral we may replace U by its reduction and assume U is reduced. This means R-invariant subschemes $W' \subset W \subset U$ of Lemma lemma-finite-flat-over-almost-dense-subscheme are equal $W' = W$. As $U = t(s^{-1}(\overline{W}))$ some point u' of the R-orbit of u is contained in $\overline{W}$ and by Lemma lemma-find-affine-integral we may replace U by $\overline{W}$ and u by u'. Hence we may assume there is a dense open R-invariant subscheme $W \subset U$ such that the morphisms s_W, t_W of the restriction (W, R_W, s_W, t_W, c_W) are finite locally free.

If $u \in W$ then we are done by Groupoids, Lemma 03JE (because W is quasi-affine so any finite set of points of W is contained in an affine open, see Properties, Lemma 01ZY). Thus we assume $u \notin W$ and hence none of the points of the orbit of u is in W. Let $\xi \in U$ be a point with a nontrivial specialization to a point u' in the orbit of u. Since there are no specializations among the points in the orbit of u (Lemma lemma-no-specializations-map-to-same-point) we see that ξ is not in the orbit. By assumption (3) we see that ξ is a generic point of U and hence $\xi \in W$. As U is

Noetherian there are finitely many of these points $\xi_1, \ldots, \xi_m \in W$. Because s_W, t_W are flat the orbit of each ξ_j consists of generic points of irreducible components of W (and hence U).

Let $j : U \to \mathrm{Spec}(A)$ be an immersion of U into an affine scheme (this is possible as U is quasi-affine). Let $J \subset A$ be an ideal such that $V(J) \cap j(W) = \emptyset$ and $V(J) \cup j(W)$ is closed. Apply Lemma lemma-find-almost-invariant-function to the groupoid scheme (W, R_W, s_W, t_W, c_W), the morphism $j|_W : W \to \mathrm{Spec}(A)$, the points ξ_j, and the ideal J to find an $f \in J$ such that $(j|_W)^{-1}D(f)$ is an R_W-invariant affine open containing ξ_j for all j. Since $f \in J$ we see that $j^{-1}D(f) \subset W$, i.e., $j^{-1}D(f)$ is an R-invariant affine open of U contained in W containing all ξ_j.

Let Z be the reduced induced closed subscheme structure on

$$U \setminus j^{-1}D(f) = j^{-1}V(f).$$

Then Z is set theoretically R-invariant (but it may not be scheme theoretically R-invariant). Let (Z, R_Z, s_Z, t_Z, c_Z) be the restriction of R to Z. Since $Z \to U$ is finite, it follows that s_Z and t_Z are finite. Since $u \in Z$ the orbit of u is in Z and agrees with the R_Z-orbit of u viewed as a point of Z. Since $\dim(\mathcal{O}_{U,u'}) \leq 1$ and since $\xi_j \notin Z$ for all j, we see that $\dim(\mathcal{O}_{Z,u'}) \leq 0$ for all u' in the orbit of u. In other words, the R_Z-orbit of u consists of generic points of irreducible components of Z.

Let $I \subset A$ be an ideal such that $V(I) \cap j(U) = \emptyset$ and $V(I) \cup j(U)$ is closed. Apply Lemma lemma-find-almost-invariant-function to the groupoid scheme (Z, R_Z, s_Z, t_Z, c_Z), the restriction $j|_Z$, the ideal I, and the point $u \in Z$ to obtain $h \in I$ such that $j^{-1}D(h) \cap Z$ is an R_Z-invariant open affine containing u.

Consider the R_W-invariant (Groupoids, Lemma 03BH) function

$$g = \mathrm{Norm}_{s_W}(t_W^\sharp(j^\sharp(h)|_W)) \in \Gamma(W, \mathcal{O}_W)$$

(In the following we only need the restriction of g to $j^{-1}D(f)$ and in this case the norm is along a finite locally free morphism of affines.) We claim that

$$V = (W_g \cap j^{-1}D(f)) \cup (j^{-1}D(h) \cap Z)$$

is an R-invariant affine open of U which finishes the proof of the lemma. It is set theoretically R-invariant by construction. As V is a constructible set, to see that it is open it suffices to show it is closed under generalization in U (Topology, Lemma 0542 or the more general Topology, Lemma 0903). Since $W_g \cap j^{-1}D(f)$ is open in U, it suffices to consider a specialization $u_1 \rightsquigarrow u_2$ of U with $u_2 \in j^{-1}D(h) \cap Z$. This means that h is nonzero in $j(u_2)$ and $u_2 \in Z$. If $u_1 \in Z$, then $j(u_1) \rightsquigarrow j(u_2)$ and since h is nonzero in $j(u_2)$ it is nonzero in $j(u_1)$ which implies $u_1 \in V$. If $u_1 \notin Z$ and also not in $W_g \cap j^{-1}D(f)$, then $u_1 \in W$, $u_1 \notin W_g$ because the complement of $Z = j^{-1}V(f)$ is contained in $W \cap j^{-1}D(f)$. Hence there exists a point $r_1 \in R$ with $s(r_1) = u_1$ such that h is zero in $t(r_1)$. Since s is finite we can find a specialization $r_1 \rightsquigarrow r_2$ with $s(r_2) = u_2$. However, then we conclude that h is zero in $u_2' = t(r_2)$ which contradicts the fact that $j^{-1}D(h) \cap Z$ is R-invariant and u_2 is in it. Thus V is open.

Observe that $V \subset j^{-1}D(h)$ for our function $h \in I$. Thus we obtain an immersion

$$j' : V \longrightarrow \mathrm{Spec}(A_h)$$

Let $f' \in A_h$ be the image of f. Then $(j')^{-1}D(f')$ is the principal open determined by g in the affine open $j^{-1}D(f)$ of U. Hence $(j')^{-1}D(f)$ is affine. Finally, $j'(V) \cap V(f') = j'(j^{-1}D(h) \cap Z)$ is closed in $\mathrm{Spec}(A_h/(f')) = \mathrm{Spec}((A/f)_h) = D(h) \cap V(f)$ by our choice of $h \in I$ and the ideal I. Hence we can apply Lemma lemma-get-affine to conclude that V is affine as claimed above. $\square$